Josh Calcino

Astrophysicist | Data Analysis | Scientific Software

josh.calcino@gmail.com | github.com/joshcalcino | Research publications

Quantitative researcher with a PhD in astrophysics. Build Python analysis pipelines, compare models with observations and quantify uncertainty; communicate results to technical and general audiences.

Technical Skills

Programming: Python (NumPy, SciPy, emcee, Numba), SQL, Git, Python–Fortran integration, AI-assisted coding.

Analysis: Bayesian inference, MCMC, likelihood analysis, AIC/BIC, Monte Carlo uncertainty propagation, numerical simulation, scientific visualization.

Experience

Postdoctoral Researcher | *Tsinghua University* Jan 2024–Present

- Lead independent research and build **Python analysis pipelines** to predict observations from simulations and **validate models against measurements**.
- Develop **DiskBridge**, including ray-tracing and chemical-modelling tools in **Python and C++, accelerated with Numba**; use AI-assisted coding daily.

Director's Postdoctoral Fellow | *Los Alamos National Laboratory* Mar 2021–Nov 2023

- Led studies **connecting analytic and hydrodynamical models with observational data**.
- Built a **data analysis pipeline** connecting hydrodynamical simulations with observed planet-forming discs; analysed **multiple terabytes of data** to derive empirical relationships.

PhD Researcher | *University of Queensland* Jul 2016–Feb 2021

- Used **Python, Bayesian inference, MCMC, and the Bayesian information criterion** to compare models of the speed at which stars orbit the Milky Way, constraining its mass and how that mass is distributed.
- Used **Monte Carlo sampling** to propagate model uncertainties and constrain how much of the Milky Way's dark matter could consist of compact objects such as black holes.
- Switched research fields to planet formation, learned the required **simulation and data analysis methods**, and published in the new field within 12 months.

Intern Software Developer | *ADACS, Swinburne University* Jun–Aug 2020

- Improved SPLASH's command-line interface and built **PySplashSPH**, a **Python–Fortran library** giving researchers direct access to simulation data and native functions from Python.
- Worked to implementation deadlines and participated in daily stand-up meetings.

Undergraduate Researcher | *University of Queensland* Mar 2015–Jul 2016

- Estimated how biases in the collection of supernovae data could bias our models that measure the expansion rate of the universe. This involved **using Python, Bayesian MCMC, and the Bayesian information criterion** to assess evidence for a bias and ascertain how such a bias could affect the recovery of other model parameters.

Communication & Leadership

- Authored **over 40 research papers, including 10 as first author**; delivered **21 research talks and seminars**, including a conference plenary.
- Delivered **six one-hour public lectures** and presented at a sold-out Pint of Science event. Co-authored an explanation of the search for extraterrestrial life for *The Conversation* and filmed a segment explaining black-hole detection for *Scope*, Network Ten's children's science program.
- Supervised **three undergraduate researchers** and chaired the organizing committee for a research school with **approximately 80 participants**.

Education

PhD, Astrophysics — University of Queensland Jul 2016–Feb 2021

First-Class Honours, Astrophysics — University of Queensland Mar 2015–Jul 2016

BSc, Physics & Applied Mathematics — Griffith University Feb 2012–Nov 2014